

C207

Data-Driven Decision Making

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A real-estate agent with 15 years of experience is describing a neighborhood that just got a new elementary school. The five homes sold there this year went for \$210,000, \$220,000, \$230,000, \$240,000 and \$1,500,000. Which figure best describes the typical sale price?

- A. The mean, \$480,000.
- B. \$225,000, the median after dropping the \$1.5 million sale.
- C. The range, \$1,290,000.
- D. The median, \$230,000.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A call center that handles about 2,000 calls a day and just bought new headsets finds that call length is normally distributed with a mean of 6 minutes and a standard deviation of 1.5 minutes. About what percentage of calls last between 3 and 9 minutes?

- A. 47.5%
- B. 99.7%
- C. 68%
- D. 95%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

Test scores are normally distributed with a mean of 50 and a standard deviation of 5. About 16% of scores lie above what value?

- A. 45
- B. 55
- C. 60
- D. 50

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A retail chain's analyst finds that, across 38 stores, the number of staff who completed customer-service training has a correlation of $r = 0.72$ with monthly sales. The training vendor is based in Chicago. A vice president concludes that training causes higher sales and wants to require it everywhere. What is the best response?

- A. An r of 0.72 is strong, so it proves training causes sales.
- B. The data show an association, not cause; larger, busier stores may both train more staff and sell more.
- C. Training explains 72% of the variation in sales.
- D. No conclusion is possible, because 38 stores is too few.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

An online retailer runs an A/B test of a new checkout page. The null hypothesis is that conversion rates are equal. The test gives $p = 0.03$ with $\alpha = 0.05$. The designer who built the new page, a recent hire from a big tech firm, is confident it works. What is the correct conclusion?

- A. Fail to reject the null, because the p-value is small.
- B. There is a 97% probability that the new page is better.
- C. Reject the null; the difference is statistically significant at the 5% level.
- D. The new page will always convert better from now on.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A test uses $\alpha = 0.01$. Which p-value leads to rejecting the null hypothesis?

- A. 0.02
- B. 0.05
- C. 0.004
- D. 0.99

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A quality team tests a sample of 400 parts from a new supplier in Mexico. The null hypothesis is that the new supplier's defect rate equals the old supplier's. The team rejects the null and drops the supplier. In truth, the two defect rates are the same. What happened?

- A. A Type I error.
- B. Sampling bias.
- C. A Type II error.
- D. No error, since dropping a supplier is always the safe choice.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A plant tests whether a new process lowers its defect rate. The null hypothesis is that it has no effect. Which outcome is a Type II error?

- A. Concluding the process has no effect when it actually lowers defects.
- B. Lowering α so Type II errors can no longer happen.
- C. Rejecting the null because p is below α .
- D. Concluding the process lowers defects when it actually has no effect.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A city wants residents' views on a proposed bus route. Staff survey people at the main downtown bus station on weekday mornings and collect 1,200 responses. The survey was designed by a university consultant. What is the main weakness of this sample?

- A. Any sample over 1,000 is always representative.
- B. 1,200 responses is too few for a city.
- C. It is a convenience sample that leaves out residents who do not ride at that time and place, so it is biased.
- D. It is a stratified random sample, so it is sound.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A bottling line's control chart has a center line of 500 ml, an upper control limit of 506 ml and a lower control limit of 494 ml. The last 20 samples fell randomly between the limits. Today a sample measured 507 ml. The filler was serviced last month by the manufacturer. What should the supervisor conclude?

- A.** A point outside the control limits signals a special cause; investigate before adjusting.
- B.** Adjust the filler after every sample that differs from 500 ml.
- C.** This is normal common-cause variation; no action is needed.
- D.** The product is out of specification and must be scrapped.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

An online hiking-gear seller estimates monthly sales with the regression equation $\text{Sales} = 12,000 + 3.5 \times \text{Ad spend}$, in dollars. The model's R^2 is 0.64, and the seller's warehouse is in Boise. If next month's ad spend is \$4,000, what are predicted sales?

- A.** \$14,000
- B.** \$26,000
- C.** \$16,640
- D.** \$12,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

Using $\text{Sales} = 12,000 + 3.5 \times \text{Ad spend}$, what ad spend predicts sales of \$33,000?

- A.** \$73,500
- B.** About \$9,429
- C.** \$21,000
- D.** \$6,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A real-estate agent ~~with 15 years of experience~~ is describing a neighborhood ~~that just got a new elementary school~~. The five homes sold there this year went for \$210,000, \$220,000, \$230,000, \$240,000 and \$1,500,000. Which figure best describes the typical sale price?

Target word: typical price **Decides:** One sale is an extreme outlier, so use the median.

A	Confusion set	\$2,400,000 / 5; the outlier drags the mean above four of the five prices.
B	Dropped element	The median already handles the outlier; deleting real data is not needed.
C	Adjacent category	Range measures spread, not a typical value.
D	Correct	The middle value is not pulled up by the \$1.5 million sale.

2. Answer D Concept 2

A call center ~~that handles about 2,000 calls a day and just bought new headsets~~ finds that call length is normally distributed with a mean of 6 minutes and a standard deviation of 1.5 minutes. About what percentage of calls last between 3 and 9 minutes?

Target word: percentage in the range **Decides:** 3 and 9 are each 2 standard deviations from 6.

A	Dropped element	Counts only one side of the mean.
B	Adjacent category	That is ± 3 SD, the range 1.5 to 10.5 minutes.
C	Confusion set	That is ± 1 SD, the range 4.5 to 7.5 minutes.
D	Correct	$(9 - 6) / 1.5 = 2$; about 95% of values fall within ± 2 SD.

3. Answer B Concept 2 · transfer

Test scores are normally distributed with a mean of 50 and a standard deviation of 5. About 16% of scores lie above what value?

Target word: cutoff for the top 16% **Decides:** 68% within ± 1 SD leaves 32% outside, 16% on each side.

A	Inverted operation	About 16% lie below 45, not above.
B	Correct	Mean + 1 SD = 55; about 16% lie above it.
C	Adjacent category	Mean + 2 SD; only about 2.5% lie above it.
D	Wrong stem number	Half the scores lie above the mean.

4. Answer B Concept 3

A retail chain's analyst finds that, ~~across 38 stores~~, the number of staff who completed customer-service training has a correlation of $r = 0.72$ with monthly sales. ~~The training vendor is based in Chicago.~~ A vice president concludes that training causes higher sales and wants to require it everywhere. What is the best response?

Target word: what the correlation supports **Decides:** Correlation alone does not establish cause.

A	Absolute language	No correlation, however strong, proves cause by itself.
B	Correct	A third variable such as store size could drive both.
C	Confusion set	Variation explained is $r^2 = 0.52$, and even that is not causal.
D	Camouflage	Answers the sample-size detail; the core flaw is the causal leap.

5. Answer C Concept 4

An online retailer runs an A/B test of a new checkout page. The null hypothesis is that conversion rates are equal. The test gives $p = 0.03$ with $\alpha = 0.05$. The designer who built the new page, ~~a recent hire from a big tech firm, is confident it works~~. What is the correct conclusion?

Target word: decision on the null **Decides:** $p = 0.03$ is less than $\alpha = 0.05$.

A	Inverted operation	A small p-value is the reason to reject.
B	Confusion set	The p-value is not the probability that a hypothesis is true.
C	Correct	$p < \alpha$ means reject the null.
D	Absolute language	Significance does not guarantee future results.

6. Answer C Concept 4 • transfer

A test uses $\alpha = 0.01$. Which p-value leads to rejecting the null hypothesis?

Target word: p-value that rejects at $\alpha = 0.01$ **Decides:** Reject only when $p < 0.01$.

A	Adjacent category	Would reject at $\alpha = 0.05$, but not at 0.01.
B	Confusion set	Mixes up the common default α with this test's α .
C	Correct	$0.004 < 0.01$.
D	Inverted operation	A large p-value gives no evidence against the null.

7. Answer A Concept 5

A quality team tests ~~a sample of 400 parts from~~ a new supplier in Mexico. The null hypothesis is that the new supplier's defect rate equals the old supplier's. The team rejects the null and drops the supplier. In truth, the two defect rates are the same. What happened?

Target word: type of error **Decides:** A true null was rejected.

A	Correct	Rejecting a null that is actually true is a Type I error.
B	Adjacent category	Bias concerns how the sample was chosen; the question is about a wrong conclusion.
C	Inverted operation	Type II is failing to reject a false null.
D	Absolute language	The decision was based on a false conclusion; that is an error.

8. Answer A Concept 5 • transfer

A plant tests whether a new process lowers its defect rate. The null hypothesis is that it has no effect. Which outcome is a Type II error?

Target word: describe a Type II error **Decides:** Type II = failing to reject a false null.

A	Correct	A real effect was missed.
B	Absolute language	Lowering α actually raises the chance of a Type II error.
C	Confusion set	That is the decision rule, not an error.
D	Inverted operation	That is a Type I error.

9. Answer C Concept 6

A city wants residents' views on a proposed bus route. Staff survey people at the main downtown bus station on weekday mornings and collect 1,200 responses. ~~The survey was designed by a university consultant.~~ What is the main weakness of this sample?

Target word: sampling weakness **Decides:** Only current weekday-morning riders at one station could be chosen.

A	Absolute language	A large biased sample is still biased.
B	Camouflage	Size is not the problem; selection is.
C	Correct	Not every resident had a chance of selection.
D	Confusion set	No strata or random selection were used.

10. Answer A Concept 7

A bottling line's control chart has a center line of 500 ml, an upper control limit of 506 ml and a lower control limit of 494 ml. The last 20 samples fell randomly between the limits. Today a sample measured 507 ml. ~~The filler was serviced last month by the manufacturer.~~ What should the supervisor conclude?

Target word: meaning of the 507 ml point **Decides:** The point is above the upper control limit.

A	Correct	Points beyond the limits are unlikely under common-cause variation alone.
B	Adjacent category	Adjusting for common-cause noise (tampering) adds variation.
C	Inverted operation	Common-cause variation stays inside the limits.
D	Confusion set	Control limits describe process behavior; specification limits come from customers.

11. Answer B Concept 8

An online hiking-gear seller estimates monthly sales with the regression equation $\text{Sales} = 12,000 + 3.5 \times \text{Ad spend}$, in dollars. ~~The model's R^2 is 0.64, and the seller's warehouse is in Boise.~~ If next month's ad spend is \$4,000, what are predicted sales?

Target word: predicted sales **Decides:** Plug \$4,000 into the equation.

A	Dropped element	Drops the intercept.
B	Correct	$12,000 + 3.5 \times 4,000 = 12,000 + 14,000 = \$26,000$.
C	Camouflage	Multiplies the prediction by R^2 , which measures fit, not output.
D	Wrong stem number	The intercept alone.

12. Answer D Concept 8 • transfer

Using $\text{Sales} = 12,000 + 3.5 \times \text{Ad spend}$, what ad spend predicts sales of \$33,000?

Target word: solve for ad spend **Decides:** $(33,000 - 12,000) / 3.5$.

A	Inverted operation	Multiplies 21,000 by 3.5 instead of dividing.
B	Dropped element	$33,000 / 3.5$ drops the intercept.
C	Adjacent category	That is the sales above the intercept, not the ad spend.
D	Correct	$21,000 / 3.5 = 6,000$.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Mean vs. median with outliers	The median resists outliers; in skewed data it describes the typical value better than the mean.	Reporting the mean when one extreme value pulls it away from most of the data.	One value far larger or smaller than the rest.
2	The empirical rule	In a normal distribution about 68% of values fall within 1 standard deviation of the mean, 95% within 2 and 99.7% within 3.	Mixing up how many standard deviations a range spans.	A mean, a standard deviation and a range or cutoff.
3	Correlation is not causation	A correlation shows that two variables move together; it does not show that one causes the other, and r^2 (not r) is the share of variation explained.	Treating a strong r as proof of cause, or reading r as a percentage.	A strong correlation followed by a causal conclusion.
4	p-values and significance	If $p < \alpha$, reject the null hypothesis; the p-value is the chance of data this extreme if the null were true, not the chance the null is true.	Reading the p-value as the probability that the result is real.	A p-value and an α in the same stem.
5	Type I and Type II errors	Type I = rejecting a true null (false alarm); Type II = failing to reject a false null (missed effect).	Swapping the two.	A conclusion and the hidden truth that contradicts it.
6	Sampling bias	A sample represents a population only if every member has a known chance of selection; convenience samples are biased no matter how large.	Believing a large sample fixes a biased selection method.	Where and how respondents were gathered.
7	Control charts	Points inside control limits with no pattern reflect common-cause variation; a point beyond the limits signals a special cause to investigate.	Adjusting a stable process after every small change, or confusing control limits with specification limits.	Upper and lower control limits and a new data point.
8	Using a regression equation	Predicted $y = \text{intercept} + \text{slope} \times x$; solve for x by subtracting the intercept and dividing by the slope.	Dropping the intercept.	An equation of the form $y = a + bx$ and a value to plug in or solve for.

Course scope (public description): This course presents critical problem-solving methodologies, including field research and data collection methods that enhance organizational performance. Topics include quantitative analysis, statistical and quality tools. You will improve your ability to use data to make informed decisions.